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Implications of cranial pinning during awake craniotomy on anesthetic requirements: A retrospective cohort study

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ABSTRACT

Background: Awake craniotomy (AC) is the gold standard for tumor resections in eloquent brain regions requiring surgical precision. Traditional AC uses pins to immobilize the head, which may contribute to scalp injury, discomfort, and hemodynamic fluctuations. We evaluated perioperative outcomes of AC performed with and without pin fixation at a single tertiary center.

Methods: We conducted a retrospective cohort study of adults undergoing AC between October 2018 and June 2023. Outcomes included head movement and movement-related workflow disruptions, anesthetic dosing, hemodynamics, operative duration, and postoperative recovery.

Results: Head movement was greater in unpinned cases ($p < 0.001$), although disruptive movements were uncommon (Grade 4: 6 %; no Grade 5 events). Propofol dosing was higher in pinned patients (3.2 ± 1.9 vs 2.4 ± 2.2 mg/kg/hr; $p = 0.029$), while dexmedetomidine dosing was similar between groups. RASS scores were comparable overall, with sex-based differences observed. Unpinned AC was associated with smaller increases in systolic blood pressure (17.5 ± 24.1 vs 25.4 ± 24.7 mmHg; $p = 0.021$), shorter operative duration (151.7 ± 56.3 vs 184.2 ± 74.7 min; $p = 0.001$), and similar ICU length of stay ($p = 0.649$).

Conclusions: Unpinned AC was associated with greater head movement but rare clinically disruptive events, alongside modest differences in anesthetic requirements, hemodynamics, and operative duration. These findings suggest potential workflow and comfort benefits in carefully selected patients rather than major safety differences. Prospective multicenter studies with standardized protocols are warranted to better define patient selection and validate these observations.

1. Introduction

Awake craniotomy (AC) is the standard for tumor resections in eloquent brain regions, allowing real-time patient feedback to preserve neurological functions [1,2]. Patients receive sedation or anesthesia during uncomfortable phases of AC, including cranial pin application, skin incision, muscle dissection, drilling, bone flap removal, and dural manipulation [3,4]. Because brain tissue lacks nociceptors, patients can remain fully conscious during mapping and resection [5]. This allows maximal tumor removal near critical areas while minimizing neurological deficits [6–9].

The Mayfield skull clamp (Integra Life Sciences, Cincinnati, OH), is commonly used to prevent head movement and ensure precision [10]. Although generally safe, pinning can cause complications such as skull fractures, epidural hematomas, or air embolism [10,11]. Their application is highly noxious and may trigger hemodynamic fluctuations [12]. Regional anesthesia, including scalp and field blocks, are routinely used to mitigate pain responses; however, they may not always fully

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prevent discomfort due to inaccurate placement or suboptimal local anesthetic (LA) dosing [13–15].

Recent advances in neuronavigation allow AC without pinning while maintaining reliable intraoperative guidance [16,17]. At our institution, optical navigation is used for pinned cases, while unpinned cases employ electromagnetic navigation (EM), improving surgical access without compromising comfort or accuracy and enabling smoother neuropsychological testing and anesthetic management [16]. At Mayo Clinic, our extensive experience with unpinned AC prompted a systematic evaluation of this approach.

This retrospective chart review evaluated perioperative differences between pinned and unpinned AC, focusing on anesthetic requirements, hemodynamic stability, head movement, operative workflow, and postoperative recovery, with the aim of assessing the clinical feasibility and safety of unpinned AC. Given the retrospective design and heterogeneous neurosurgical population, this study aimed not to evaluate a single AC technique, but to characterize real-world practice patterns and perioperative associations related to cranial pinning within an established multidisciplinary AC program. These findings may help inform strategies to optimize patient outcomes and enhance perioperative safety and comfort.

2. Methods

2.1. Study design

This retrospective cohort study included data from October 1, 2018, to June 30, 2023, and was Institutional Review Board (IRB)-exempt (IRB #23–002468; April 27, 2023), with a waiver of Health Insurance Portability and Accountability Act (HIPAA) authorization. The study adhered to Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guidelines and was conducted in accordance with established standards [18].

2.2. Patient selection

We reviewed the electronic medical records of 223 adults who underwent AC at Mayo Clinic Florida for indications including gliomas, metastatic lesions, vascular malformations, and epilepsy resections, with no exclusions.

2.3. Procedure

At our institution, AC is performed within a standardized, multidisciplinary program involving neurosurgeons, neuroanesthesiologists, neurologists, and clinical neuropsychologists, as previously described [19]. All patients received a preoperative scalp block targeting the supraorbital, supratrochlear, zygomaticotemporal, auriculotemporal, and greater and lesser occipital nerves using standard institutional techniques [3]. LA type and total dose were individualized based on patient characteristics and procedural requirements, consistent with established practice. Mayfield clamp use was determined by surgeon preference and case-specific factors, including lesion location, complexity, and procedural requirements. As a result, pinning status was not randomized. In the OR, anesthesia was delivered per standard practice; most patients received moderate sedation for bone flap removal and dural exposure, while others remained lightly sedated and conversational. Propofol and dexmedetomidine were administered alone or in combination at the anesthesiologist's discretion and titrated to balance comfort, immobility during noxious phases, and responsiveness during mapping; boluses were used as needed and were included in total anesthetic dosing. All patients were continuously monitored with standard ASA monitors, including pulse oximetry, ECG, noninvasive blood pressure, and end-tidal CO_2 via nasal cannula or facemask.

Sedation was paused during cortical mapping and neurologic assessment [3]. Neuronavigation was used in all cases for incision planning, surgical guidance, and assessment of resection extent. Pinned cases used optical navigation with Mayfield-mounted reference arrays, while unpinned cases used electromagnetic navigation with scalp-based arrays. Real-time neuropsychological testing was performed during resection to monitor cognition and preserve function [20–22], and electroencephalography was used to detect stimulation-induced after-discharges and seizures [23]. Sedation was resumed during wound closure for comfort. Total surgery time was defined from incision to wound closure, so operative duration in pinned cases included additional steps required for pinning. Postoperatively, all patients were transferred to the intensive care unit (ICU) for monitoring.

2.4. Outcomes

The primary endpoint was intraoperative anesthetic use. A standardized anesthesia protocol was followed, with sedatives and analgesics titrated to ensure patient comfort, immobility during critical phases, responsiveness during mapping, and hemodynamic stability. Drug selection and dosing were individualized but consistent with institutional practice, and anesthetic data, including LA doses, were extracted from anesthesia charts. Secondary endpoints included intraoperative head movement, total opioid use, hemodynamic changes, pain scores, Richmond Agitation Sedation Scale (RASS) [24] assessments, and ICU length of stay.

Head movement was evaluated by a trained rater (W.C.) blinded to study objectives. Patients were recorded using an iPad-mounted camera, and videos from cortical mapping and neuropsychological testing were reviewed. Each evaluable video segment was graded on a 6-point Likert scale: 0 = no visible movement; 1 = micro movement (imperceptible without magnification); 2 = slight movement (e.g., talking or respiration); 3 = noticeable, nondisruptive movement (did not disrupt workflow or require correction, e.g., deep respiration or coughing); 4 = noticeable, disruptive movement (interrupted workflow and required correction without threatening patient safety); and 5 = emergent, disruptive movement (significant head shift, lift, or turn requiring immediate action due to threatened patient safety). An arithmetic mean head movement score was calculated for each patient, and the distribution of grades was expressed as the proportion of trials per rating. A second rater (D.S.S.) independently evaluated a random subset (n = 10) to assess scoring consistency. Inter-rater reliability was good (weighted Cohen's $\kappa = 0.77$), consistent with standards for ordinal scales [25,26].

Electronic medical records were reviewed to determine total opioid use, including hydromorphone, oxycodone, fentanyl, hydrocodone, and tramadol, accounting for scheduled doses, rescue medications, and IV patient-controlled analgesia (IV-PCA). All opioid doses were converted to morphine milligram equivalents (MMEs) using the formula: *MME/day = strength/unit x units/day x MME conversion factor* [27–29]. Non-opioid analgesics were routinely given postoperatively per institutional practice; their use did not differ meaningfully between pinned and unpinned cohorts and was not analyzed separately.

Hemodynamic data were collected at baseline (pre-incision) and at the point of maximum change. Pain was assessed using the numerical rating scale at multiple time points up to 24 h postoperatively. RASS assessments were performed repeatedly during the first 24 h, with values other than zero (alert and calm) included in the analysis. Peripheral oxygen saturation (SpO_2) was recorded from baseline to the lowest observed value. Exploratory analyses evaluated potential differences by gender and relevant comorbidities to further assess patient responses.

2.5. Statistical analysis

Analyses comparing unpinned and pinned cohorts were performed using SAS 9.4 and JMP 17.0 (Cary, NC). Descriptive statistics characterized the study sample, and statistical significance was set at $p < 0.05$

with 95 % confidence intervals. Hypothesis testing was applied to primary and secondary outcomes, while comparisons by gender and comorbidities were exploratory. Continuous variables (e.g., LA dose, surgery time, ICU stay, heart rate, and systolic blood pressure changes) were summarized as mean ± SD or median (IQR) based on distribution. Categorical variables (e.g., sex, ASA score, RASS, comorbidities) were reported as counts and percentages.

Continuous variables were visually assessed for normality using histograms and quantile-quantile (Q-Q) plots stratified by pinning status. Given the moderate sample size and clinical data, group comparisons used Welch's two-sample *t*-test for its robustness to unequal variances and mild deviations from normality. Nonparametric tests, such as the Mann-Whitney *U* test, were applied to clearly non-normally distributed or ordinal outcomes, including head movement scores.

Anesthetic dosing was analyzed separately. Total intraoperative propofol and dexmedetomidine doses (infusions and boluses) were weight-normalized and divided by total surgery time to calculate mg/kg/hr for propofol and μg/kg/hr for dexmedetomidine. Cases with missing data or no drug administration were excluded from drug-specific analyses. Group means were compared using two-sided Welch's *t*-tests.

Univariate and multivariate analyses were performed to compare cohorts. Exploratory decision tree models and ordinal logistic regression examined associations between pinning status and key clinical outcomes, with patient-level covariates (age, sex, comorbidity burden, repeat surgery) treated as potential effect modifiers rather than for confirmatory adjustment. Decision tree analyses used G-square and Logworth tests. Quantitative variables (e.g., age, BMI) were compared using Welch's *t*-test, and categorical variables (e.g., procedure side, location) were assessed with chi-square tests to confirm cohort comparability.

3. Results

3.1. Demographics

A total of 223 ACs performed between October 1, 2018, and June 30, 2023, were included, with 140 (62.8 %) using pin fixation and 83 (37.2 %) without. Patient demographics are summarized in Table 1. Mean age was 50.9 ± 16.4 years in the pinned group and 52.1 ± 16.0 years in the unpinned group. The pinned cohort comprised 84 males (60 %) and 56 females (40 %), while the unpinned cohort included 51 males (61.4 %) and 32 females (38.6 %). Exploratory analysis of common comorbidities (diabetes, hypertension, hyperlipidemia, hypothyroidism, and obesity) revealed no statistically significant differences between groups. In the pinned cohort, lesions were 43.6 % frontal, 7.9 % insular, 3.6 % multifocal, 1.4 % occipital, 12.1 % parietal, and 30.7 % temporal. In the unpinned cohort, lesions were 46.9 % frontal, 7.2 % insular, 1.2 % multifocal, 0 % occipital, 13.3 % parietal, and 31.3 % temporal.

Table 1
Baseline demographic characteristics by cohort (pinned vs unpinned).

Characteristic	Pinned (n = 140)	Unpinned (n = 83)
Age, yrs (mean ± SD)	50.9 ± 16.4	52.1 ± 16.0
Gender, n (%)		
Male	84 (60.0)	51 (61.4)
Female	56 (40.0)	32 (38.6)
BMI, kg/m^2 (mean ± SD)	27.2 ± 5.4	28.3 ± 5.3
ASA class, n (%)		
I-II	38 (27.1)	23 (27.7)
III-IV	102 (72.9)	60 (72.3)
Comorbidities, n (%)		
Hypertension	30 (21.4)	24 (28.9)
Diabetes mellitus	7 (5.0)	5 (6.0)
Hyperlipidemia	18 (12.9)	18 (21.7)
Obesity	37 (26.4)	27 (32.5)
Hypothyroidism	9 (6.4)	7 (8.4)
No. of Comorbidities, n (%)		
0	83 (59.3)	32 (38.6)
1	30 (21.4)	29 (34.9)
2	15 (10.7)	15 (18.1)
3	7 (5.0)	6 (7.2)
4	5 (3.6)	1 (1.2)
Race, n (%)		
White	123 (87.9)	73 (87.9)
Black	5 (3.6)	2 (2.4)
Asian	6 (4.3)	2 (2.4)
Native Americans	6 (4.3)	6 (7.2)
Ethnicity, n (%)		
Hispanic	20 (14.3)	13 (15.7)
Non-Hispanic	114 (81.4)	66 (79.5)
Lesion location, n (%)		
Frontal	61 (43.6)	39 (46.9)
Temporal	43 (30.7)	26 (31.3)
Parietal	17 (12.1)	11 (13.3)
Occipital	2 (1.4)	0 (0.0)
Insular	12 (8.6)	6 (7.2)
Multifocal	5 (3.6)	1 (1.2)

Abbreviations: ASA, American Society of Anesthesiologists Physical Status Classification; BMI, Body Mass Index; SD, Standard Deviation; kg/m^2, kilograms per square meter; yrs, years.

3.2. Total anesthetic dose (Primary outcome)

Propofol was administered in 62 pinned and 41 unpinned cases. Among recipients, the mean propofol dose was 3.2 ± 1.9 mg/kg/hr in pinned cases and 2.4 ± 2.2 mg/kg/hr in unpinned cases (difference 0.8 mg/kg/hr, 95 % CI 0.1–1.4; p = 0.029) (Supplementary Figure 1a). Dexmedetomidine was given in 110 pinned and 73 unpinned cases, with mean doses of 0.6 ± 0.3 μg/kg/hr in pinned and 0.6 ± 0.4 μg/kg/hr in unpinned cases (difference −0.1 μg/kg/hr, 95 % CI −0.2–0.0; p = 0.205) (Supplementary Figure 1b).

3.3. Local anesthetic dose

In exploratory analyses, pinning status was not a primary predictor of LA dose, and differences by pinning were not statistically significant. LA dose was therefore analyzed primarily by comorbidity burden, where it increased with greater comorbidity count, with secondary, non-significant differences observed by pinning status. Patients with four comorbidities required a larger dose (195.8 ± 57.9 mg) than those with three or fewer comorbidities (139.6 ± 59.4 mg). Among patients with three or fewer comorbidities, pinned patients required less LA (133.6 ± 61.6 mg) than unpinned patients (149.9 ± 54.4 mg; t = 2.35, p = 0.060). Analyses stratified by comorbidity were exploratory (Supplementary Figure 2).

3.4. Head movement

Head movement represented the most prominent differentiating outcome between cohorts. Mean head movement grade per video trial was significantly greater in unpinned cases compared with pinned cases (0.6 ± 0.6 vs 0.4 ± 0.5, respectively). Overall head movement scores differed significantly between groups, with greater movement observed in unpinned cases (Mann–Whitney U = 1616.5, z = −3.53, p < 0.001; rank-biserial correlation r ≈ 0.29) [30].

Video data were available for 147 patients (94 pinned, 53 unpinned); missing data were due to absent recordings or patients being out of camera view. The number of evaluable video trials per patient did not differ between groups (pinned: 141.7 ± 108.8; unpinned: 131.3 ± 92.0; p = 0.832). A total of 19,251 video trials were analyzed, including 12,326 from pinned cases and 6925 from unpinned cases. Most trials in both groups demonstrated no observable head movement (Grade 0). Disruptive movements were rare, with Grade 4 movements observed in 8 of 6925 trials (0.12 %) in the unpinned group and in none of the 12,326 trials in the pinned group. No Grade 5 emergent movements

were observed in either cohort. At the patient level, Grade 4 disruptive movements occurred in 3 of 53 unpinned cases (6 %) and in 0 of 94 pinned cases (0 %). Most observed movements were classified as Grades 1–3 and did not require corrective action.

3.5. Total postoperative opioid use

Postoperative opioid consumption was numerically higher in the pinned cohort compared with the unpinned cohort (8.4 ± 9.8 vs 7.0 ± 10.8 MME/day; $p = 0.431$). A similar pattern was observed during sex-stratified exploratory analysis with higher mean opioid use observed in pinned compared with unpinned female patients (8.63 ± 9.4 vs 6.45 ± 7.9 MME/day), although this difference was not statistically significant ($p = 0.350$) (Supplementary Figure 3).

3.6. Richmond Agitation-Sedation Scale (RASS)

Overall, there were no significant differences in postoperative sedation between pinned and unpinned cohorts. RASS < 0 scores were similar (pinned 2.7 ± 4.6 vs unpinned 2.9 ± 4.6; $p = 0.751$), and RASS > 0 scores were also comparable (pinned 0.3 ± 1.2 vs unpinned 0.3 ± 0.8; $p = 0.890$). In exploratory sex-stratified analyses of RASS < 0, females had comparable scores regardless of pinning (3.47 ± 5.35). Among males, unpinned patients had significantly lower RASS < 0 scores than pinned patients (1.73 ± 2.81 vs 2.82 ± 4.42; $t = 2.51$, $p < 0.01$) (Supplementary Figure 4).

3.7. Baseline and absolute intraoperative variables

Baseline and absolute intraoperative values for SBP, DBP, HR, and SpO_2 are summarized in Table 2. Baseline values were similar between pinned and unpinned groups, and no clinically meaningful differences were observed in absolute intraoperative maxima or minima. For this reason, subsequent analyses emphasized changes from baseline, which better capture intraoperative physiologic responses while accounting for inter-individual variability.

3.8. Systolic blood pressure

The change in SBP from baseline to the intraoperative maximum differed between groups, with the pinned cohort demonstrating a greater mean increase (25.4 ± 24.7 mmHg) compared with unpinned cases (17.5 ± 24.1 mmHg; $p = 0.021$). Exploratory analyses stratified by first versus repeat surgery did not demonstrate statistically significant differences (14.3 ± 21.6 mmHg vs 22.3 ± 27.1 mmHg; $t = 1.42$, $p < 0.160$) (Supplementary Figure 5).

3.9. Heart rate

The mean change in heart rate from baseline to intraoperative maximum was numerically higher in the pinned cohort (16.4 ± 20.0 bpm) compared with the unpinned cohort (12.3 ± 19.7 bpm), although this difference did not reach statistical significance ($p < 0.137$) (Supplementary Figure 6).

Table 2
Baseline and absolute intraoperative changes in hemodynamic and respiratory variables by pinning status.

Variable	Pinned (Mean ± SD)	Unpinned (Mean ± SD)
Absolute change in high SBP to baseline (mmHg)	27.5 ± 22.3	22.8 ± 19.1
Absolute change in high DBP to baseline (mmHg)	12.9 ± 12.7	15.0 ± 15.1
Absolute change in baseline HR to low HR (bpm)	22.5 ± 16.6	25.1 ± 15.9
Absolute change in baseline SpO_2 to low SpO_2 (%)	6.3 ± 6.0	4.9 ± 4.7

Values are presented as mean ± standard deviation (SD). Absolute changes represent the difference between baseline and intraoperative maximum or minimum values as indicated. Abbreviations: SBP, systolic blood pressure; DBP, diastolic blood pressure; HR, heart rate; SpO_2, peripheral oxygen saturation.

3.10. Oxygen saturation

The mean decrease in SpO_2 from baseline to the lowest recorded value was numerically greater in the pinned cohort (6.0 ± 6.3 %) compared with the unpinned cohort (4.6 ± 5.1 %), although this difference did not reach statistical significance ($p < 0.075$) (Supplementary Figure 7).

3.11. Operative duration and ICU stay

Mean total surgery time differed between groups, with a shorter duration observed in the unpinned cohort (151.7 ± 56.3 min) compared with the pinned cohort (184.2 ± 74.7 min; $p < 0.001$) (Supplementary Figure 8). ICU length of stay was similar between both groups (median 1 day [IQR 1–2]; $p = 0.649$).

3.12. Pain scores and other outcomes

No significant differences were observed in pain scores (moderate 4–6, severe 7–10), RASS > 0, other HR variations, or diastolic blood pressure (DBP) changes.

All outcomes described above are summarized by cohort in Table 3.

4. Discussion

In this retrospective cohort study comparing pinned and unpinned AC, the most clinically relevant difference between fixation strategies was head movement. Although unpinned cases demonstrated higher mean movement scores, absolute differences were small, with disruptive movements requiring correction occurring in only 6 % of cases and no safety-threatening events observed. These findings suggest that, in carefully selected patients, unpinned techniques may be feasible, while highlighting head movement as the principal trade-off clinicians must consider when selecting fixation strategies.

Acceptable movement during AC likely depends on lesion location, surgical phase, and surgeon risk tolerance. In our cohort, pinning was favored for cases requiring maximal immobility, including a higher proportion of insular and multifocal lesions. Accordingly, the low rate of disruptive movement observed in unpinned cases may not generalize to more complex procedures. Although inadvertent movement poses theoretical safety concerns, no neurological injury, navigation loss, or emergent intervention occurred. These findings therefore demonstrate feasibility and safety in selected patients within an experienced program rather than universal endorsement of unpinned techniques.

AC is commonly performed with cranial fixation to optimize surgical precision [31,32], but pin application can trigger pain and sympathetic activation [33]. In our study, pinning was associated with higher propofol dosing, consistent with greater nociceptive burden requiring deeper hypnosis to maintain comfort and immobility. This is pharmacologically expected, as propofol provides rapid, titratable hypnosis and effective movement suppression, whereas dexmedetomidine primarily offers sedation and analgesia without reliably inhibiting movement at standard doses [34–36]. Dexmedetomidine was used similarly across groups and functioned as an adjunct rather than a primary hypnotic. At our institution, remifentanil is rarely used during AC due to airway and titration concerns [37]; instead, propofol with dexmedetomidine is used to balance sedation and patient cooperation [19].

Although statistically significant, absolute differences in propofol dosing were modest and within routine clinical titration ranges. These differences did not translate into clinically meaningful safety concerns, but rather reflect differences in anesthetic burden and workflow associated with fixation strategy. We therefore focused on cumulative anesthetic exposure over operative time rather than peak infusion rates,

Table 3
Primary and secondary outcomes by cohort (pinned vs unpinned).

Outcome	Pinned	Unpinned	p-value
Primary outcome			
Total anesthetic dose			
Propofol dose, mg/kg/hr (mean ± SD)[a]	3.2 ± 1.9	2.4 ± 2.2	0.029
Dexmedetomidine dose, μg/kg/hr (mean ± SD)[a]	0.6 ± 0.3	0.6 ± 0.4	0.205
Secondary outcomes			
Local anesthetic dose, mg (mean ± SD)[b]	133.6 ± 61.6	149.9 ± 54.4	0.060
Head movement[c]			
Number of movements (mean ± SD)[c]	0.4 ± 0.5	0.6 ± 0.6	0.001
Grade 4 disruptive movements, n (%)[c]	0/94 (0)	3/53 (6)	-
Grade 5 disruptive movements, n (%)[c]	0/94 (0)	0/53 (0)	-
Total opioid use, MME/day (mean ± SD)	8.4 ± 9.8	7.0 ± 10.8	0.431
Pain score 4–6 (mean ± SD)	2.6 ± 2.8	3.0 ± 3.2	0.354
Pain score 7–10 (mean ± SD)	2.1 ± 3.4	1.9 ± 3.9	0.841
RASS Score (mean ± SD)[b]			
RASS < 0	2.7 ± 4.6	2.9 ± 4.6	0.751
RASS > 0	0.3 ± 1.2	0.3 ± 0.8	0.890
Hemodynamics (mean ± SD)			
SBP change (baseline to max), mmHg	25.4 ± 24.7	17.5 ± 24.1	0.021
DBP change (baseline to max), mmHg	3.3 ± 17.9	4.4 ± 20.9	0.676
HR change (baseline to max), bpm	16.4 ± 20.0	12.3 ± 19.7	0.137
SpO_2 change (baseline to min), %	6.0 ± 6.3	4.6 ± 5.1	0.075
Total surgery time, mins (mean ± SD)	184.2 ± 74.7	151.7 ± 56.3	0.001
ICU length of stay, days (median [IQR])	1 (1–2)	1 (1–2)	0.649

Data are presented as mean ± SD, median [IQR], or n (%), as appropriate.

[a] Propofol administered in 41 unpinned and 62 pinned cases; dexmedetomidine administered in 73 unpinned and 110 pinned cases.

[b] Secondary stratified exploratory analyses are presented in Supplementary Figures 1.

[c] Head movement analyses were limited to cases with available intraoperative video recordings (n = 147; 94 pinned and 53 unpinned). Due to the low frequency of Grade 4 and absence of Grade 5 movements, formal statistical testing was not performed to generate a p-value. Abbreviations: bpm, beats per minute; CI, confidence interval; DBP, diastolic blood pressure; HR, heart rate; ICU, intensive care unit; IQR, interquartile range; max, maximum; min, minimum; mins, minutes; mmHg, millimeters of mercury; MME, morphine milligram equivalents; RASS, Richmond Agitation–Sedation Scale; SBP, systolic blood pressure; SD, standard deviation; SpO_2, peripheral capillary oxygen saturation.

as maximum doses are highly context-dependent and difficult to compare meaningfully across AC cases. Bolus-related physiologic effects were not analyzed separately due to multifactorial influences and limitations of retrospective attribution.

Respiratory depression with propofol during AC is a recognized concern. In our cohort, propofol was used at low, titrated infusion rates consistent with accepted practice and typically paused during mapping. Continuous end-tidal CO_2 monitoring enabled early detection of hypoventilation or apnea, and no cases of unrecognized respiratory compromise, emergent airway intervention, or procedure abandonment were observed. Although transient effects may be underreported in retrospective data, the absence of clinically significant sequelae supports the overall safety of this approach in a monitored multidisciplinary setting.

Headache, or pin-site pain, is a known concern with fixation devices. Prior studies report variable rates of intraoperative discomfort and supplementary analgesic requirements [38–40]. In our study, mean postoperative opioid consumption was higher in the pinned cohort, particularly among females, though differences were not statistically significant. Scalp block is a cornerstone of AC, reducing pain, sympathetic activation, and sedative requirements while facilitating patient participation during mapping [41,42]. All patients in this study received a standardized scalp block, which likely enhanced overall safety and operative conditions but does not explain the between-group differences observed.

Despite scalp block and moderate sedation, cranial pin application can elicit autonomic responses, with prior studies reporting increases in HR and SBP [43]. In our study, pinned cases demonstrated greater mean SBP changes than unpinned cases, consistent with a higher nociceptive and stress burden associated with fixation. However, these differences were modest in absolute magnitude and did not result in clinically meaningful instability or require additional pharmacologic intervention. HR and SpO_2 differences between groups were small and not statistically significant. These findings underscore the importance of distinguishing statistical significance from clinical relevance, as perioperative safety and the need for corrective interventions may better reflect the practical impact of physiologic changes during AC.

The growing use of EM navigation may help mitigate patient discomfort and limitations of head fixation, while facilitating airway access [44]. In our study, navigation modality was inherently linked to fixation strategy; therefore, differences in operative duration or workflow may partly reflect navigation technology independent of head fixation. Accordingly, efficiency differences cannot be attributed solely to pinning.

Total operative time was shorter in unpinned cases, likely reflecting elimination of pin placement, reduced need for anesthetic adjustment, and workflow differences. Lesion complexity and surgeon preference also contributed, as pinned cases more often involved insular or multifocal lesions, which independently prolong operative duration. These differences are therefore multifactorial rather than solely attributable to fixation strategy.

This study is limited by its retrospective, single-center design and potential confounding by indication, as pinning was determined by surgeon preference and case complexity rather than randomization. The pinned cohort included a higher proportion of insular and multifocal lesions, which are independently associated with longer operative times and greater anesthetic requirements; accordingly, observed differences in duration and anesthetic dosing cannot be attributed solely to pinning. Head movement ratings were not independently validated, and raters were not fully blinded to pinning status, introducing potential observer bias. Volumetric tumor data were unavailable, which may affect interpretation of operative duration and anesthetic needs. Given residual confounding from unmeasured factors, these findings should be considered hypothesis-generating. Prospective, multicenter studies with standardized protocols and direct measures of patient comfort and workflow impact are warranted.

5. Conclusion

This study provides observational insights into the impact of pinning during AC, highlighting potential associations with perioperative management. The most prominent difference between cohorts was greater head movement in unpinned cases, although clinically disruptive or safety-threatening events were rare. Unpinned surgeries were associated with lower anesthetic requirements, smaller hemodynamic changes, and shorter operative durations, while observed physiologic and anesthetic differences were statistically significant but modest in absolute magnitude. Together, these findings suggest potential workflow and comfort benefits rather than major safety differences. Overall, these results highlight trade-offs clinicians may consider when selecting fixation strategies and suggest that unpinned AC may be feasible in selected cases. Prospective multicenter studies with standardized protocols are warranted to better define patient selection and further explore patient-centered outcomes.

CRediT authorship contribution statement

Kaisorn L. Chaichana: Writing – review & editing, Supervision,

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT (OpenAI) to assist with language editing and sentence refinement. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Appendix A. Supporting information

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Data Availability

The datasets generated and analyzed during this study are not publicly available due to HIPAA-protected patient privacy. They can be obtained from the corresponding author upon reasonable request and subject to institutional approvals.

References

[1] M.S. Berger, Functional mapping-guided resection of low-grade gliomas, Clin. Neurosurg. 42 (1995) 437–452.
[2] M.S. Berger, G.A. Ojemann, Intraoperative brain mapping techniques in neuro-oncology, Stereo Funct. Neurosurg. 58 (1-4) (1992) 153–161.
[3] C.I. Eseonu, K. ReFaey, O. Garcia, A. John, A. Quinones-Hinojosa, P. Tripathi, Awake craniotomy anesthesia: a comparison of the monitored anesthesia care and asleep-awake-asleep techniques, World Neurosurg. 104 (2017) 679–686.
[4] P. Suarez-Meade, L. Marenco-Hillembrand, C. Prevatt, R. Murguia-Fuentes, A. Mohamed, T. Alsaeed, E.J. Lehrer, T. Brigham, H. Ruiz-Garcia, D. Sabsevitz, E. H. Middlebrooks, P.S. Bechtle, A. Quinones-Hinojosa, K.L. Chaichana, Awake vs. asleep motor mapping for glioma resection: a systematic review and meta-analysis, Acta Neurochir. (Wien.) 162 (7) (2020) 1709–1720.
[5] L.M. Terrier, D. Fontaine, Intracranial nociception, Rev. Neurol. (Paris) 177 (7) (2021) 765–772.
[6] R.A. Danks, L.S. Aglio, L.D. Gugino, P.M. Black, Craniotomy under local anesthesia and monitored conscious sedation for the resection of tumors involving eloquent cortex, J. Neurooncol. 49 (2) (2000) 131–139.
[7] A. Groshev, D. Padalia, S. Patel, R. Garcia-Getting, S. Sahebjam, P.A. Forsyth, F. D. Vrionis, A.B. Etame, Clinical outcomes from maximum-safe resection of primary and metastatic brain tumors using awake craniotomy, Clin. Neurol. Neurosurg. 157 (2017) 25–30.
[8] M.D. Taylor, M. Bernstein, Awake craniotomy with brain mapping as the routine surgical approach to treating patients with supratentorial intraaxial tumors: a prospective trial of 200 cases, J. Neurosurg. 90 (1) (1999) 35–41.
[9] J.O. Ekert, D.S. Sabsevitz, I. Martin Del Campo, A. Goyal, C.S. Gillespie, E. H. Middlebrooks, K.L. Chaichana, K.S. Lee, J.E. Sanchez-Garavito, A. Quinones-Hinojosa, Awake brain mapping paradigms for nondominant hemisphere gliomas, Neurosurg. Focus 56 (2) (2024) E7.
[10] P.A. Beuriat, T. Jacquesson, E. Jouanneau, M. Berhouma, Headholders' - complications in neurosurgery: a review of the literature and recommendations for its use, Neurochirurgie 62 (6) (2016) 289–294.
[11] D. Thijs, T. Menovsky, The mayfield skull clamp: a literature review of its complications and technical nuances for application, World Neurosurg. 151 (2021) 102–109.
[12] C.H. Koo, S. Jeon, J. Kim, J.H. Ryu, The effects of intravenous dexmedetomidine on hemodynamic response in patients undergoing skull-pin head-holder application during neurosurgery - a meta-analysis of randomized controlled trials, Clin. Neurol. Neurosurg. 195 (2020) 105939.
[13] A. Papangelou, B.R. Radzik, T. Smith, A. Gottschalk, A review of scalp blockade for cranial surgery, J. Clin. Anesth. 25 (2) (2013) 150–159.
[14] E. Bojaxhi, C. Louie, K. ReFaey, S.E. Gruenbaum, B.J. Leone, P. Bechtel, M. P. Barbosa, K.L. Chaichana, A. Quinones-Hinojosa, Reduced pain and opioid use in the early postoperative period in patients undergoing a frontotemporal craniotomy under regional vs general anesthesia, World Neurosurg. 150 (2021) e31–e37.
[15] A. Agarwal, P.K. Sinha, C.M. Pandey, A. Gaur, C.K. Pandey, S. Kaushik, Effect of a subanesthetic dose of intravenous ketamine and/or local anesthetic infiltration on hemodynamic responses to skull-pin placement: a prospective, placebo-controlled, randomized, double-blind study, J. Neurosurg. Anesth. 13 (3) (2001) 189–194.
[16] E. Harwick, I. Singhal, B. Conway, W. Mueller, R. Treffy, M.O. Krucoff, Pinless electromagnetic neuronavigation during awake craniotomies: technical pearls, pitfalls, and nuances, World Neurosurg. 175 (2023) e159–e166.
[17] C. Hayhurst, P. Byrne, P.R. Eldridge, C.L. Mallucci, Application of electromagnetic technology to neuronavigation: a revolution in image-guided neurosurgery, J. Neurosurg. 111 (6) (2009) 1179–1184.
[18] E. von Elm, D.G. Altman, M. Egger, S.J. Pocock, P.C. Gotzsche, J. P. Vandenbroucke, S. Initiative, The strengthening the reporting of observational studies in epidemiology (STROBE) statement: guidelines for reporting observational studies, J. Clin. Epidemiol. 61 (4) (2008) 344–349.
[19] D. Moniz-Garcia, E. Bojaxhi, B.J. Borah, R. Dholakia, H. Kim, B. Sousa-Pinto, J. P. Almeida, M. Mendhi, W.D. Freeman, W. Sherman, L. Christel, S. Rosenfeld, S. S. Grewal, E.H. Middlebrooks, D. Sabsevitz, B.F. Gruenbaum, K.L. Chaichana, A. Quinones-Hinojosa, Awake craniotomy program implementation, JAMA Netw. Open 7 (1) (2024) e2352917.
[20] B. Tomasino, T. Ius, M. Skrap, Brain Mapping: Real-Time Neuropsychological Testing Experience during Low-Grade Tumor Resection, in: W. Debinski (Ed.), Gliomas, Brisbane (AU), 2021.
[21] M. Skrap, D. Marin, T. Ius, F. Fabbro, B. Tomasino, Brain mapping: a novel intraoperative neuropsychological approach, in: J Neurosurg, 125, 2016, pp. 877–887.
[22] H. Ruiz-Garcia, E. Middlebrooks, D. Sabsevitz, P. Bechtle, A. Quinones-Hinojosa, Mapping the Brain During Awake Surgery for Patients with Neurological Disorders Including Brain Tumors, Principles of Neuro-Oncology: Brain & Skull Base, Springer, 2020, pp. 413–436.
[23] B.E. Freund, A.M. Feyissa, A. Khan, J.I. Sirven, S.S. Grewal, D. Sabsevitz, D. Moniz-Garcia, A. Quinones-Hinojosa, W.O. Tatum, Enhanced sensitivity of electrocorticography during awake craniotomy using a novel circular grid electrode, J. Neuro-Oncol. 165 (2) (2023) 313–320.
[24] C.N. Sessler, M.S. Gosnell, M.J. Grap, G.M. Brophy, P.V.O. Neal, K.A. Keane, E. P. Tesoro, R.K. Elswick, The richmond agitation-sedation scale: validity and reliability in adult intensive care unit patients, Am. J. Respir. Crit. Care Med. 166 (10) (2002) 1338–1344.
[25] S. Vanbelle, C.H. Engelhart, E. Blix, A comprehensive guide to study the agreement and reliability of multi-observer ordinal data, BMC Med. Res. Methodol. 24 (1) (2024) 310.
[26] J.R. Landis, G.G. Koch, The measurement of observer agreement for categorical data, Biometrics 33 (1) (1977) 159–174.
[27] J. Fudin, M. Raouf, E.L. Wegrzyn, M.E. Schatman, Safety concerns with the centers for disease control opioid calculator, J. Pain. Res. 11 (2018) 1–4.
[28] J.A. Lossio-Ventura, W. Song, M. Sainlaire, P.C. Dykes, T. Hernandez-Boussard, Opioid2MME: Standardizing opioid prescriptions to morphine milligram equivalents from electronic health records, Int. J. Med. Inf. 162 (2022) 104739.
[29] A. Goud, E. Kiefer, M.S. Keller, L. Truong, S. SooHoo, R.V. Riggs, Calculating maximum morphine equivalent daily dose from prescription directions for use in the electronic health record: a case report, JAMIA Open 2 (3) (2019) 296–300.
[30] C.O. Fritz, P.E. Morris, J.J. Richler, Effect size estimates: current use, calculations, and interpretation, J. Exp. Psychol. Gen. 141 (1) (2012) 2–18.
[31] I.J. Barrenechea, H. Rojas, M. Nicola, L. Marquez, R. Herrera, F. Van Isseldyk, A novel temporary cranial fixation device for awake cranial surgery: technical report of 14 cases, Surg. Neurol. Int. 11 (2020).
[32] P.-A. Beuriat, T. Jacquesson, E. Jouanneau, M. Berhouma, Headholders'–complications in neurosurgery: a review of the literature and recommendations for its use, Neurochirurgie 62 (6) (2016) 289–294.
[33] D. Thijs, T. Menovsky, The mayfield skull clamp: a literature review of its complications and technical nuances for application, World Neurosurg. 151 (2021) 102–109.

[34] D. Viderman, F. Nabidollayeva, F. Bilotta, Y.G. Abdildin, Comparison of dexmedetomidine and propofol for sedation in awake craniotomy: a meta-analysis, Clin. Neurol. Neurosurg. 226 (2023) 107623.
[35] N. Goettel, S. Bharadwaj, L. Venkatraghavan, J. Mehta, M. Bernstein, P. H. Manninen, Dexmedetomidine vs propofol-remifentanil conscious sedation for awake craniotomy: a prospective randomized controlled trial, Br. J. Anaesth. 116 (6) (2016) 811–821.
[36] S.-l Shen, J.-y Zheng, J. Zhang, W.-y Wang, T. Jin, J. Zhu, Q. Zhang, Comparison of dexmedetomidine and propofol for conscious sedation in awake craniotomy: a prospective, double-blind, randomized, and controlled clinical trial, Ann. Pharmacother. 47 (11) (2013) 1391–1399.
[37] A.E. Elbakry, E. Ibrahim, Propofol-dexmedetomidine versus propofol-remifentanil conscious sedation for awake craniotomy during epilepsy surgery, Minerva Anestesiol. 83 (12) (2017) 1248–1254.
[38] J.C. Keifer, D. Dentchev, K. Little, D.S. Warner, A.H. Friedman, C.O. Borel, A retrospective analysis of a remifentanil/propofol general anesthetic for craniotomy before awake functional brain mapping, Anesth. Analg. 101 (2) (2005) 502–508.
[39] I.J. Barrenechea, H. Rojas, M. Nicola, L. Marquez, R. Herrera, F. Van Isseldyk, A novel temporary cranial fixation device for awake cranial surgery: technical report of 14 cases, Surg. Neurol. Int. 11 (2020) 12.
[40] K.J.O. Khu, J.S.G. Pascual, K.H.D. Ignacio, Patient-reported intraoperative experiences during awake craniotomy for brain tumors: a scoping review, Neurosurg. Rev. 45 (5) (2022) 3093–3107.
[41] T. Duda, M. Lannon, P. Gandhi, A. Martyniuk, F. Farrokhyar, S. Sharma, Systematic review and meta-analysis of randomized controlled trials for scalp block in craniotomy, Neurosurgery 93 (1) (2023) 4–23.
[42] M. Luo, X. Zhao, M. Deng, Y. Hu, X. Yang, Z. Mei, L. Meng, Y. Wang, Scalp nerve block, local anesthetic infiltration, and postoperative pain after craniotomy: a systematic review and network meta-analysis of randomized trials, J. Neurosurg. Anesthesiol. 35 (4) (2023) 361–374.
[43] J.W. Potters, M. Klimek, Local anesthetics for brain tumor resection: current perspectives, Local Reg. Anesth. 11 (2018) 1–8.
[44] A.A. Morsy, W.H. Ng, Awake craniotomy using electromagnetic navigation technology without rigid pin fixation, J. Clin. Neurosci. 22 (11) (2015) 1827–1829.